

<C Programming and Lab> Chapter 2 Variables and Data Types Lab Problems

※ Notes on the Problems

- In the output examples, □ denotes a blank space that is printed at the very beginning and at the very end of each line.
- In the input/output examples, the text that follows ↪ is an explanation of the corresponding input or output
- The sample submit function may be used on the OJ (Online Judge)

[Problem 1] Write a program that reads two integers and computes their sum.

- In the input example, the **underlined bold text** indicates the values entered by the user.
- □ denotes an [Enter] key press
- The same convention applies to all the lab problems below.

Input Example 1	Output Example 1
<u>4</u> □ <u>5</u> □	sum: 9

Input Example 2	Output Example 2
<u>-12345</u> □ <u>5</u> □	sum: -12340

[Problem 2] Declare two floating-point variables, initialize them as shown below, and then write a program that computes the sum of the two numbers.

- Initial values of the floating-point numbers: 4.200000 and 5.100000
- Declaring the variables as float introduces rounding error, so declare them as double.

Input Example 1	Output Example 1
	first number: 4.200000 second number: 5.100000 sum: 9.300000

[Problem 3] Write a program that reads an integer from the user and stores it in a variable of int type, copies that value into a variable of char type, and prints both values using the %d conversion specification.

- Run the program with the input values given below. (Understand the range of representable values of each data type.)
- 10, -10, 100, -100, 200, -200, 500, -500

Input Example 1	Output Example 1
<u>10</u> □	(int->int) 10 (int->char->int) 10

Input Example 2

200

Output Example 2

```
(int->int) 200
(int->char->int) -56
```

- A variable ch of char type can store only values between -128 and 127. If 200 is assigned to ch, the value exceeds 127 so it cannot be stored correctly. Therefore the incorrect value -56 is stored instead.

[Problem 4] Write a program that reads an integer from the user and stores it in a variable of int type. Rewrite the previous problem using a variable of unsigned char type instead of a variable of char type, and run it with the same input values. (Understand the range of representation of the data types.)

Input Example 1

200

Output Example 1

```
(int->int) 200
(int->unsigned char->int) 200
```

Input: Explain why the output differs between [Problem 3] and [Problem 4] for the input 200

(Answer) A variable ch of char type can store only values between -128 and 127.

A variable ch of unsigned char type can store values between 0 and 255.

[Problem 5] Print the following on the screen. (Use escape sequence)

Input Example 1 Output Example 1

	The double quotation mark is ". The single quotation mark is '. Character A is 'A'. To represent a double quotation mark, use the escape sequence \".
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[Problem 6] Read a digit between 0 and 9 using the conversion specification '%c', and then write the code that prints the same value as shown below using the conversion specifications '%c' and '%d'. (An application of understanding what a character really is.)

- In order to print the value as an integer, use the fact that the ASCII code values of the digits '0' through '9' are consecutive.
- You do not need to know the ASCII code value of each digit.

Input Example 1

4

Output Example 1

```
(%c) 4
(%d) 4
```

In the last line of the example, convert the character '4' into integer 4.

e.g.) character '4' - character '0' = 52 in the ASCII table - 48 in the ASCII table = integer 4

- To print a % character on the screen, write two % characters; that is printf("%%");

[Problem 7] The ASCII code values of the English lower-case letters a-z are consecutive in

alphabetical order. Using this property, write a program that prints the position of a lower-case letter entered by the user within the alphabet.

- Assume that a is the 0th letter and z is the 25th letter.
- You do not need to know the ASCII code value of each character.

Input Example 1 Output Example 1

b	b is the 1 th lower case letter in the English alphabet.
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Input Example 2 Output Example 2

y	y is the 24 th lower case letter in the English alphabet.
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[Problem 8] Write a program that reads an integer between 0 and 25 and prints the English upper-case letter located at the corresponding position.

- Assume that the 0th letter is A and the 25th letter is Z.
- You do not need to know the ASCII code value of each character.

Input Example 1 Output Example 1

1	The 1 th upper case letter in the English alphabet is B.
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Input Example 2 Output Example 2

24	The 24 th upper case letter in the English alphabet is Y.
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[Problem 9] Write a program that reads an English lower-case letter and converts it into an upper-case letter.

- Hint) Combine Problem 7 and Problem 8.

Input Example 1 Output Example 1

b	B
---	---

Input Example 2 Output Example 2

y	Y
---	---

[Problem 10] Conversely to the previous problem, write a program that reads an English upper-case letter and converts it into a lower-case letter.

Input Example 1 Output Example 1

B	b
---	---

Input Example 2 Output Example 2

Y	y
---	---